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Future extreme precipitation amplified by intensified mesoscale moisture convergence

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Extreme precipitation events are driven by complex multiscale atmospheric dynamic interactions, fuelled by available moisture. They are expected to intensify with climate change, posing increasing risks to human communities and ecosystems. However, current low-resolution climate models struggle to accurately represent key extreme precipitation-generating phenomena, limiting our ability to generate robust and reliable future projections. Here we present an ensemble of climate simulations with a 10-to-25-km resolution and an improved representation of mesoscale convective systems to assess future changes in daily extreme precipitation and its drivers. Our high-resolution simulations more realistically capture the observed spatial distribution and intensity of daily extreme precipitation over the historical period than the 100-km resolution counterparts. In a future scenario with high carbon dioxide emissions, daily extreme precipitation over land could increase by about 41% by 2100, mainly as a result of increased mesoscale moisture convergence. The impact of this dynamical contribution to extreme precipitation is underestimated by a factor of three in the low-resolution model. These results highlight the crucial role of high-resolution climate modelling in constraining future extremes and informing more effective climate risk assessments and adaptation strategies.

As climate change accelerates, extreme precipitation continues to intensify, consistent with theoretical considerations[1–4] and projections from the Coupled Model Intercomparison Project Phases 5 and 6 (CMIP5 and CMIP6)[5,6]. Early studies emphasized thermodynamics as the primary driver of extreme precipitation changes, with the Clausius–Clapeyron (CC) relationship predicting an ~7% K^{-1} increase in extreme precipitation with temperature[1,2]. While CC-scaling provides a baseline for projections, recent studies reveal a more complex relationship between extreme precipitation and temperature[7,8], influenced by atmospheric dynamics, particularly changes in the strength and location of updrafts and local flow convergence[5,9–15]. This dependence on atmospheric dynamics poses challenges for CMIP-class models, which parameterize small-scale updrafts due to their coarse resolution (~100 km), introducing uncertainties in simulating extreme precipitation[16]. Updraft dynamics involve intricate, multiscale interactions among atmospheric phenomena[17].

Kilometre-scale resolution models capable of explicitly representing deep-convective processes can better resolve the atmospheric dynamics driving extreme precipitation[18]. However, because of computational constraints, such models have been limited to regional studies[19]. While coupled global kilometre-scale or storm-resolving Earth system models are under development, century-long climate

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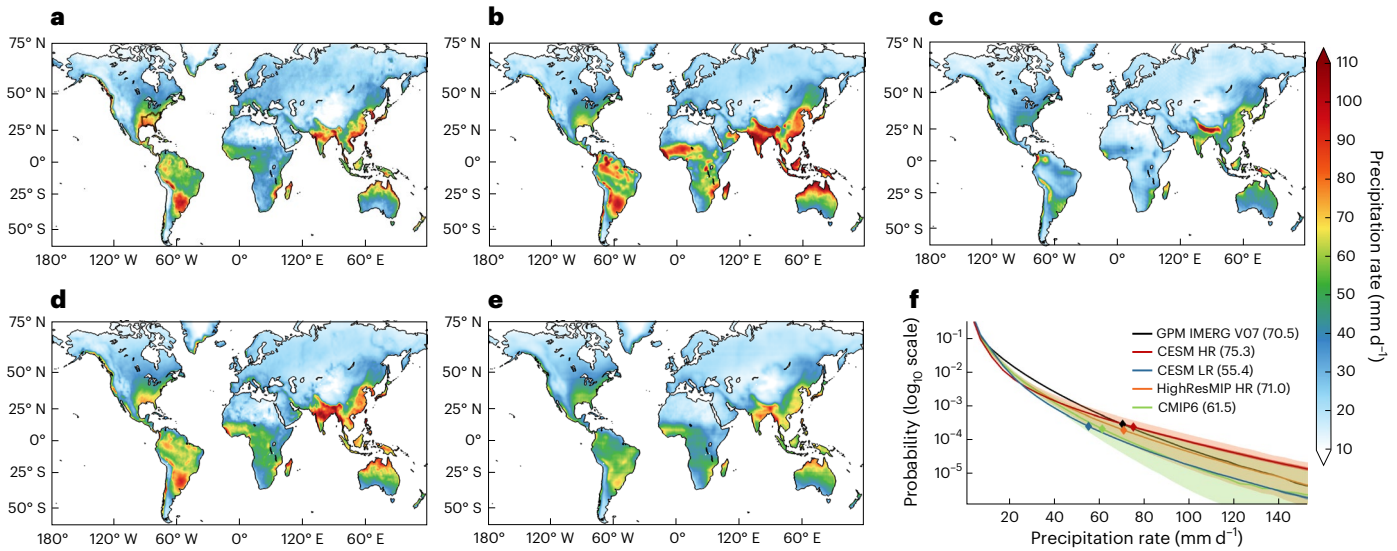


Fig. 1 | Observed and simulated annual R99p and PDFs. a, Observed R99p (mm d^{-1}) over the 1998–2024 period from IMERG V07. The brown contour outlines the SEUS region, an area of intense extreme precipitation. **b**,**c**, As in **a**, but for the ten-member ensemble mean of CESM-HR (**b**) and CESM-LR (**c**). **d**,**e**, As in **a**, but for the nine-model multimodel mean of HighResMIP (**d**) and 35-model multimodel mean of CMIP6 simulations (**e**) for the period 1998–2014. **f**, Global-land-averaged precipitation PDFs for the different models. Shading shows ensemble spread (minimum–maximum range) and the diamond-shaped dots and numbers indicate precipitation values at the 99.9th percentile. To ensure fair comparison, all data are first regridded to common 1° × 1° grid using conservative interpolation method.

simulations remain a scientific and technical challenge[20–23]. Efforts have therefore focused on global high-resolution models (10–25 km), which bridge the gap between CMIP-class (~100 km) and storm-resolving (~1 km) models. Even with this improved resolution, computational demands still limit large ensembles of full-length historical-and-future climate simulations under standard CMIP protocols[24].

Despite these limitations, many studies show that moving from ~100 km- to 10–25 km-resolution enhances the simulated extreme precipitations, particularly in mid- and high-latitudes, where synoptic-scale frontal systems are better captured[25–28]. However, a detailed understanding of the processes driving these improvements and a systematic evaluation of high-resolution versus CMIP-class model future climate projections remains incomplete, raising a critical question: can high-resolution models better capture extreme precipitation drivers globally compared with CMIP-class models and provide more reliable projections of future extreme precipitation changes?

We address this question using a new suite of historical-and-future climate simulations[29] featuring a high-resolution (~25 km) atmosphere-and-land model coupled with an eddy-rich (~10 km) ocean and sea-ice model, based on the Community Earth System Model v.1.3 (CESM-HR; Methods). Unlike previous efforts[24,30,31], this suite includes ten-member high-resolution ensembles spanning 1920–2100, following the CMIP5 protocol under the representative concentration pathways (RCP) 8.5 and 6.0 scenario. These simulations are supported by the understanding the role of MESoscale Atmosphere-Ocean Interactions in Seasonal-to-Decadal CLImate Prediction (MESACLIP) and Improving Prediction and Projection Of Gulf of Mexico Sea-Level Changes using Eddy-Resolving Earth System Models (iPOGS) projects (Methods). Leveraging the newly developed Multi-Object Analysis of Atmospheric Phenomenon (MOAAP; Methods), these new ensembles reveal a key advancement: CESM-HR, despite not storm-resolving, demonstrates a step-change improvement in the spatiotemporal representation of mesoscale phenomena compared with its low-resolution counterpart (CESM-LR). By realistically simulating multiscale interactions, particularly between mesoscale convective systems (MCSs) and other atmospheric phenomena, CESM-HR better captures the dynamic drivers of extreme precipitation, enabling more reliable global future projections. Towards the end of the century, CESM-HR projects substantially greater increases in extreme precipitation compared with CESM-LR, as a result of enhanced mesoscale moisture convergence, with consistent results across all members and robust uncertainty quantification.

Observed and simulated extreme precipitation

The improvements in CESM-HR over CESM-LR are striking when comparing simulated and satellite-observed extreme precipitation using accumulated precipitation from days exceeding the 99th percentile of wet-day precipitation (R99p), computed on a common 1° × 1° coarsened grid with the conservative interpolation method (Fig. 1). CESM-HR, although slightly overestimating, more accurately captures the global distribution and R99p amplitude, with a global mean of 41.3 mm d^{-1}, closer to the observed value of 38.3 mm d^{-1}, compared with 28.1 mm d^{-1} for CESM-LR. The pattern correlation with observations is 0.91 for CESM-HR (Fig. 1a,b) versus 0.83 for CESM-LR (Fig. 1a,c) and CESM-HR reduces the globally averaged root-mean-square error of R99p by nearly 20% (12.8 mm d^{-1}) over CESM-LR (16.1 mm d^{-1}).

Comparing observed and simulated extreme precipitation intensities is subject to considerable uncertainty as a result of observational limitations, methodological sensitivities and metric choices[32]. Previous studies have shown that spatial interpolation methods can introduce uncertainties when comparing extreme precipitation[33]. To assess the sensitivity of our comparison results to interpolation choices, we repeat our R99p analyses using a bilinear interpolation method (Supplementary Fig. 1; Methods). Consistent with earlier findings[25,28], bilinear interpolation increases R99p in both observations and simulations. Nevertheless, the overall conclusion remains unchanged: CESM-HR captures extreme precipitation more accurately than CESM-LR (Supplementary Fig. 1). This result is further supported by a comparison of annual maximum daily precipitation (Rx1day). Unlike R99p, Rx1day is based on annual maxima and thus requires long-term records for robust sampling. We use observed Rx1day derived from station data extending back to the early twentieth century as a reliable reference for model evaluation. When comparing observed and simulated Rx1day on the observational grid using

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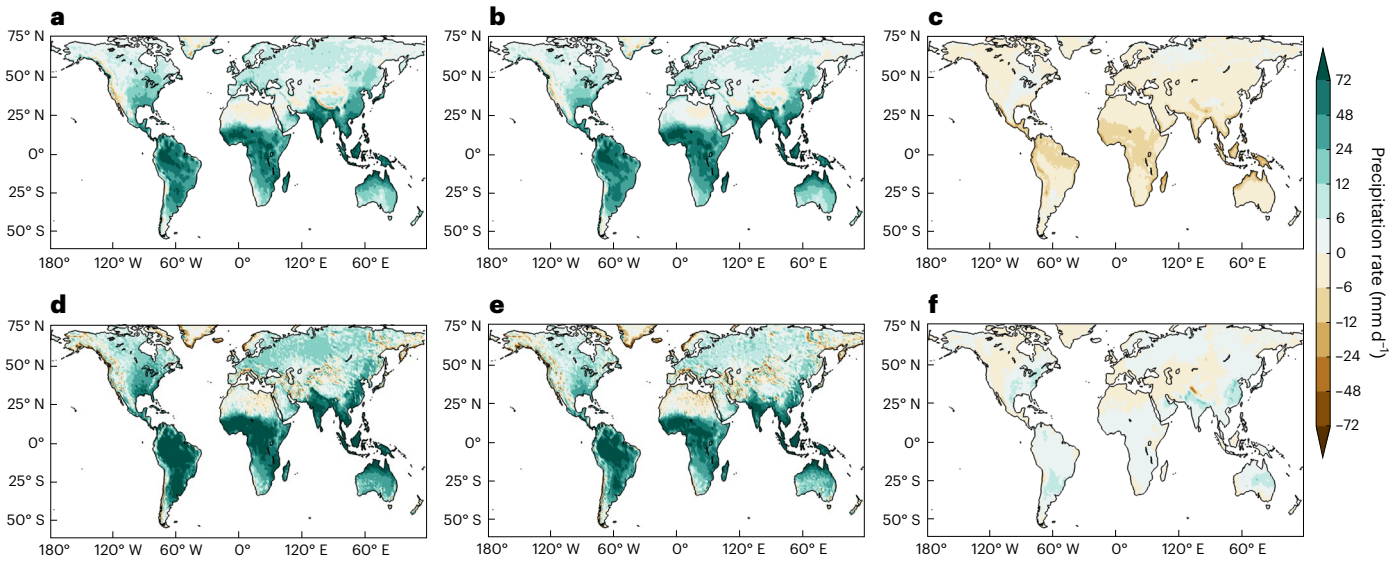


Fig. 2 | Moisture budget for Rx1day differences: CESM-HR − CESM-LR. **a**, Simulated mean Rx1day (mm d^{-1}) for the 1981–2010 period in CESM-HR minus that in CESM-LR. **b**, As in **a**, but for the resolved precipitation component. **c**, As in **a**, but for the parameterized precipitation component. **d**, Rx1day difference (mm d^{-1}) between CESM-HR and CESM-LR driven by differences in moisture flux convergence component related to horizontal flow convergence (M_c). **e**, As in **d**, but driven by $M_{\Delta C}$, which is the component of M_c related to the horizontal flow convergence differences between CESM-HR and CESM-LR. **f**, As in **d**, but driven by $M_{\Delta q}$, which is the component of M_c related to the moisture differences between CESM-HR and CESM-LR.

bilinear interpolation, CESM-HR again shows clear improvements over CESM-LR (Extended Data Fig. 1). Thus, regardless of interpolation method, observational dataset or extreme precipitation metric, CESM-HR provides a more realistic representation of extreme precipitation intensities.

The improved representation of extreme precipitation with enhanced model resolution is evident not only in CESM but also in other CMIP models. Comparing the multimodel ensemble mean (MMEM) R99p from HighResMIP simulations to the CMIP6 ensemble reveals similar improvements, with a global-land-averaged MMEM R99p of 38.8 mm d^{-1} for HighResMIP versus 34.2 mm d^{-1} for CMIP6 (Fig. 1d,e). Global-land-region daily precipitation probability distribution functions (PDFs) show that higher resolution better captures heavy precipitation rates above 40 mm d^{-1}, aligning more closely with observations (Fig. 1f). At the 99.9th percentile, precipitation rates are 75.3 mm d^{-1} in CESM-HR and 71.0 mm d^{-1} in HighResMIP, much closer to the observed 70.5 mm d^{-1}, compared with 55.4 mm d^{-1} in CESM-LR and 61.5 mm d^{-1} in CMIP6.

These improvements extend to historical trends, where CESM-HR more accurately captures both the magnitude and distribution of observed increases in extreme precipitation over recent decades. For the trend comparison and subsequent model analyses, we use Rx1day, derived from long observations and model records for more robust statistics than R99p from the short satellite record. On regional scales, CESM-HR more accurately captures the observed increasing trends in western and northern Australia, Scandinavia, South Africa and Southeast China than does CESM-LR, although both models fail to reproduce the observed decreases over Chile, western Argentina and Bolivia (Supplementary Fig. 2). While comparing observed and simulated trends involves uncertainties, as observed trends are influenced by internal climate variability whereas simulated trends primarily capture externally forced responses due to ensemble averaging, these results underscore the superiority of high-resolution models in simulating both mean and long-term changes in extreme precipitation.

Extreme precipitation driver differences

Previous studies show that increasing model horizontal resolution enhances extreme precipitation simulations by increasing resolved large-scale (larger than grid size) precipitation and reducing parameterized subgrid-scale (smaller than grid size) precipitation[26,29,34,35]. Our analysis supports these findings, revealing that the Rx1day increase in CESM-HR is primarily driven by resolved precipitation increase, especially in regions with intense extreme precipitation (Fig. 2a,b). This increase is partially offset by a reduction in parameterized precipitation relative to CESM-LR (Fig. 2c). Globally, resolved and parameterized precipitation contribute 92% and 8%, respectively, to mean Rx1day in CESM-HR, compared with 74% and 26% in CESM-LR.

The greater contribution from resolved precipitation in CESM-HR suggests a better capture of small-scale circulations, enhancing flow convergence and updraft strength[34]. To confirm that flow convergence differences primarily drive Rx1day differences between CESM-HR and CESM-LR, we conducted a moisture budget analysis[13,36,37]. Moisture flux convergence, a dominant process in the moisture budget, can be broken down into contributions from horizontal flow convergence (M_c) and moisture advection (M_a)[36]. For extreme precipitation events, substantial cancellation occurs between M_a and the moisture tendency (M_t), while evaporation (E) is minimal, leaving M_c as the dominant factor[13,38]. Our analysis reveals that Rx1day differences between CESM-HR and CESM-LR are primarily driven by M_c differences, with spatial patterns and magnitude closely matching (pattern correlation of 0.8; Fig. 2a,d). Differences in other moisture budget processes are much smaller (Supplementary Fig. 3).

Further decomposition of M_c differences reveals that the dynamical tendency ($M_{\Delta C}$), driven by differences in dry flow convergence, is the primary driver of Rx1day differences, with the thermodynamical tendency ($M_{\Delta q}$), related to moisture profile differences, playing a minor role (Methods; Fig. 2e,f). These findings suggest that CESM-HR can better capture small-scale circulation and atmospheric dynamics, resulting in stronger flow convergence and improved updraft strength, which are the primary factors behind the Rx1day increase in CESM-HR. This underscores the importance of improving model dynamics through increasing resolution to improve extreme precipitation simulations. Next, we will further demonstrate how improved dynamics in CESM-HR lead to better simulations of atmospheric phenomena and their multiscale interactions driving extreme precipitation events.

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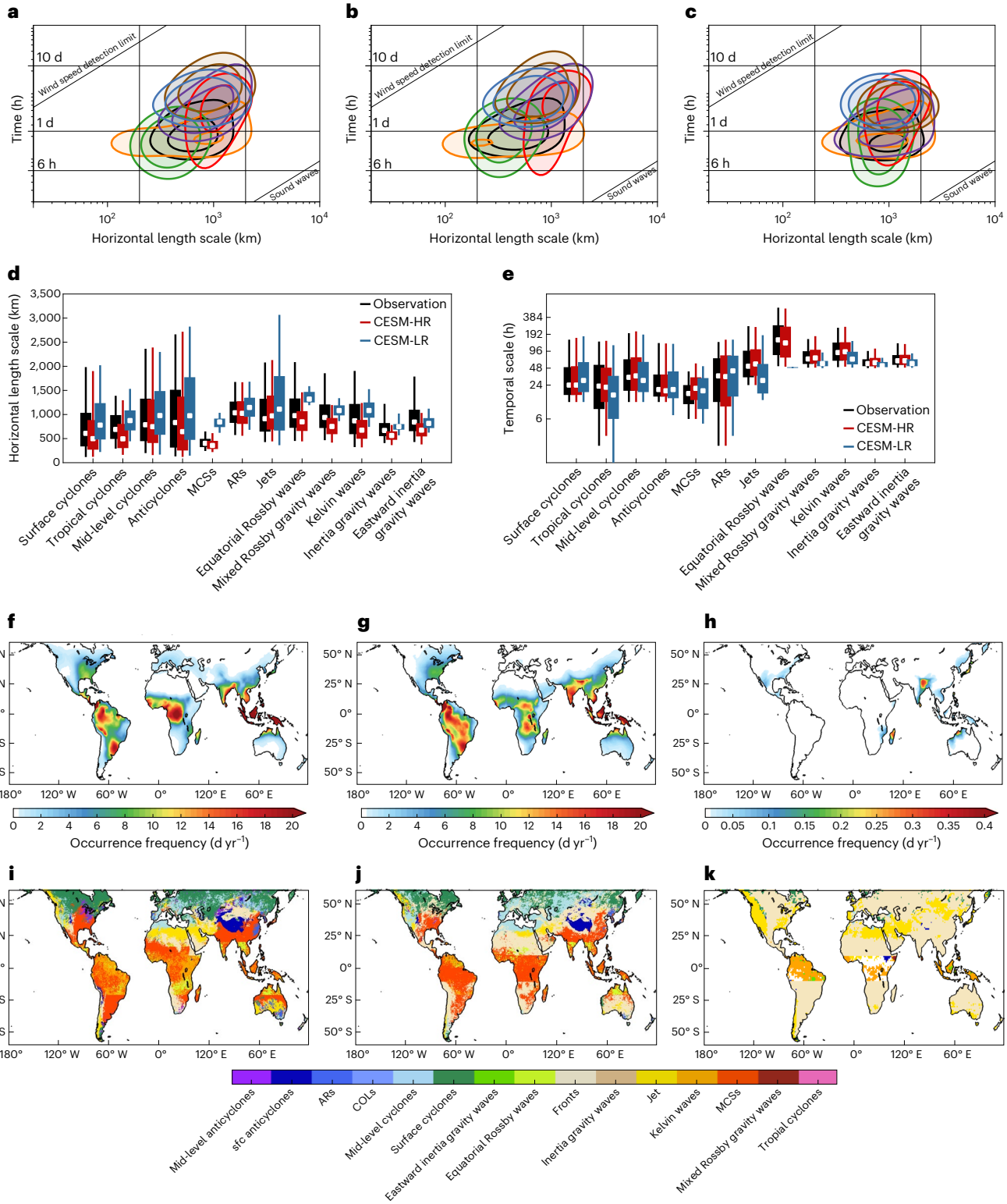


Fig. 3 | Extreme precipitation-producing atmospheric phenomena tracked by MOAAP programme. a, Scale diagram of the tracked phenomena from the observations, with spatial scales (km) on the *x* axis and time scales (h) on the *y* axis for cyclones (black), anticyclones (orange), MCSs (green), ARs (red), jets (purple) and westward-propagating (brown) and eastward-propagating (blue) equatorial waves. Contours show two-dimensional Gaussian kernel density estimates with a bandwidth of 0.4. **b**, Same as **a**, but from CESM-HR. **c**, Same as **a**, but from CESM-LR. **d**, Box-and-whisker plot of spatial scales for each tracked phenomenon, showing the median (white dot), interquartile range (boxes) and 5th to 95th percentiles (whiskers) across observations, CESM-HR and CESM-LR. The sample sizes used to construct these box-and-whisker plots are provided in Supplementary Table 3. **e**, Same as **d**, but for timescales of the tracked phenomena. **f**–**h**, Occurrence frequency (d yr^{-1}) of MCSs over global land for observations (**f**), CESM-HR (**g**) and CESM-LR (**h**), with the colour scale in **h** for CESM-LR reduced by eight times compared with observations and CESM-HR. **i**–**k**, The most frequent occurring atmospheric phenomenon detected by the MOAAP algorithm during Rx1day events over global land for observations (**i**), CESM-HR (**j**) and CESM-LR (**k**). White areas indicate regions where no tracked phenomenon is associated with Rx1day events.

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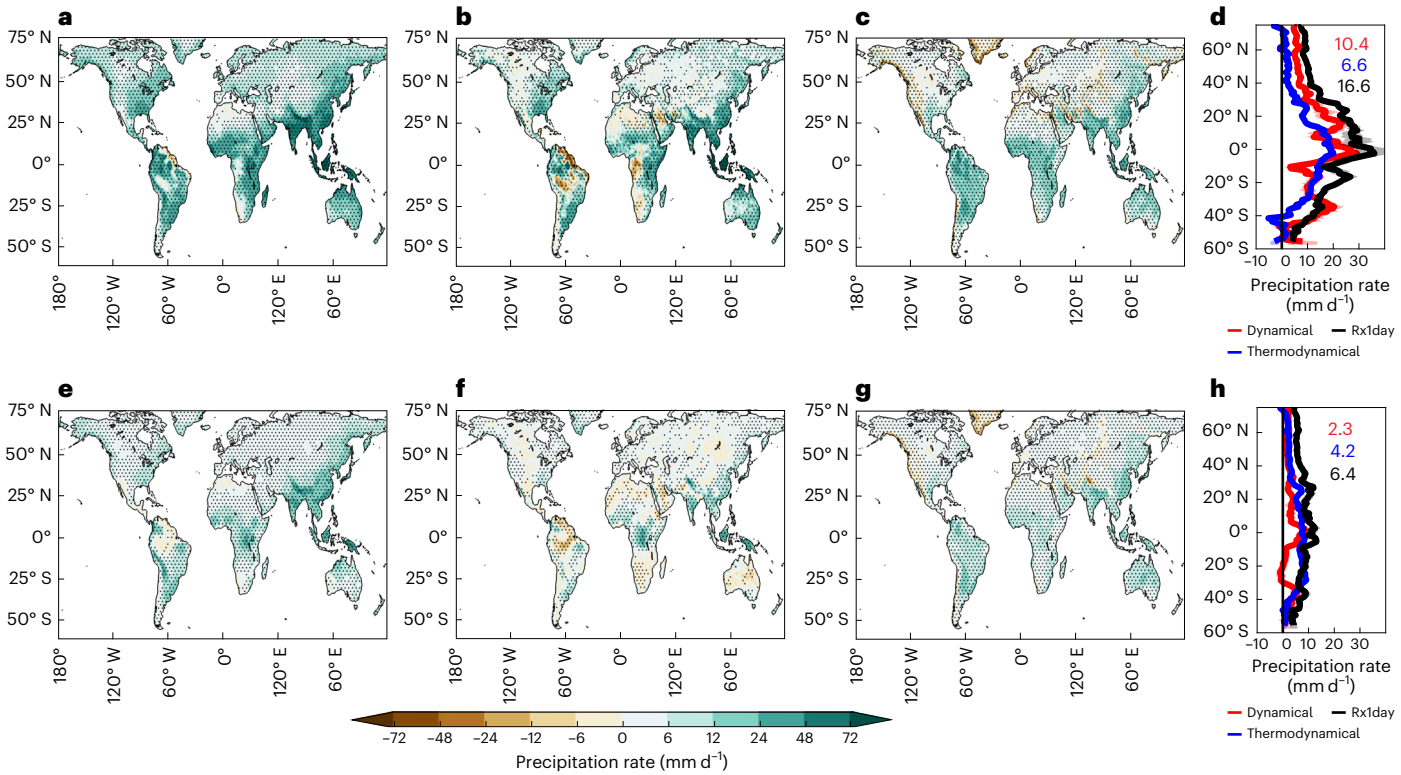


Fig. 4 | Projected Rx1day changes and their dynamical versus thermodynamical drivers. a, Future changes in Rx1day (mm d^{-1}) over global-land-averaged for 2071–2100, relative to the 1981–2010 period, in CESM-HR. **b**, Future Rx1day changes driven by dynamical tendency in CESM-HR. **c**, As in **b**, but for thermodynamical tendency (Methods). **d**, Zonally averaged Rx1day changes (16.6) driven by dynamical (10.4) and thermodynamical (6.6) tendency across global land for CESM-HR. Shading represents the standard deviation across ten-member ensembles. The area-weighted global-land-averaged values are listed. Owing to the presence of a small residual in the moisture budget analysis, the sum of the dynamical and thermodynamical tendency terms does not exactly equal the total Rx1day changes. **e**–**g**, As in **a** (**e**), **b** (**f**) and **c** (**g**), but for CESM-LR. **h**, Zonally averaged Rx1day changes (6.4) driven by dynamical (2.3) and thermodynamical (4.2) tendency across global land for CESM-LR. Stippling indicates regions where three-quarters of the ensemble members agree on the sign of the ensemble mean.

Precipitation-generating atmospheric phenomena and their multiscale interactions

Extreme precipitation events are often linked to mesoscale processes triggered, enhanced or organized by larger-scale phenomena, underscoring the importance of multiscale interactions[39]. Using MOAAP (Methods), we identify and track key atmospheric phenomena driving extreme precipitation in CESM-HR and CESM-LR and compare them to observations from a previous study[17] using the fifth generation ECMWF atmospheric reanalysis (ERA5)[40] combined with the integrated multisatellite retrievals for global precipitation measurement (IMERG V07)[41] and merged infrared (GPM-MERGIR)[42] data. This comparison evaluates how model resolution affects the simulation of extreme precipitation-producing phenomena and their interactions. We note that MOAAP feature tracking requires hourly outputs, which are only available for one member of CESM-HR and CESM-LR ensemble. However, given the large number of tracked features in this single member, expanding the ensemble size is unlikely to notably impact the results.

The tracked phenomena include MCSs, tropical cyclones (TCs), anticyclones, mid-level cyclones and cut-off lows (COLs), atmospheric rivers (ARs), jet streams, frontal systems and various types of atmospheric waves[17] (Methods). A comparison of scale diagrams shows that CESM-HR closely replicates the observed spatial and temporal scales for these phenomena (Fig. 3a,b), whereas CESM-LR generally shifts towards larger spatial and shorter temporal scales (Fig. 3a,c), indicating difficulty in capturing small-scale systems like MCSs, which are key drivers of extreme precipitation. This difference is further illustrated in spatial- and temporal-scale box-whisker plots (Fig. 3d,e).

Occurrence frequencies of key tracked phenomena reveal even starker differences. Specifically, while CESM-HR reproduces the observed distribution and frequency of MCSs, closely aligning with the Rx1day pattern, CESM-LR shows almost no MCSs (Fig. 3f–h). Instead, CESM-LR exhibits an excessive activity of jets and fronts compared with both observations and CESM-HR (Extended Data Fig. 2). These discrepancies impact how the models represent extreme precipitation drivers. To illustrate, we identify the most frequent phenomenon occurring with Rx1day at each global-land-grid point (Fig. 3i–k). In both observations and CESM-HR, MCSs are the phenomenon most often linked to Rx1day events in regions where these events are intense (Fig. 1). By contrast, CESM-LR associates Rx1day almost exclusively with fronts and jets (Fig. 3k), probably because of its coarse resolution-limiting multiscale interactions. MCS development requires sufficient moisture, instability, a lifting mechanism for convection initiation and vertical wind shear to organize storms into larger systems[43], conditions supported by concurrent atmospheric phenomena that facilitate MCS development through multiscale interactions[17].

We illustrate these processes by focusing on the southeast USA (SEUS), a region of intense extreme precipitation (Fig. 1a), where MCSs are the most frequent phenomenon associated with Rx1day events in both observations and CESM-HR (Fig. 3i,j). During 2001–2020, MCSs account for 26.5% of Rx1day events in observations and 29.2% in CESM-HR, but only about 0.5% in CESM-LR (Extended Data Fig. 3). The next most frequent phenomena in observations and CESM-HR are fronts (18.2% and 27.7%, respectively) and ARs (14.8% and 12.6%). By stark contrast, CESM-LR shows fronts as the most dominant phenomena (32.1%), followed by jets (25.2%) and surface cyclones (13.1%).

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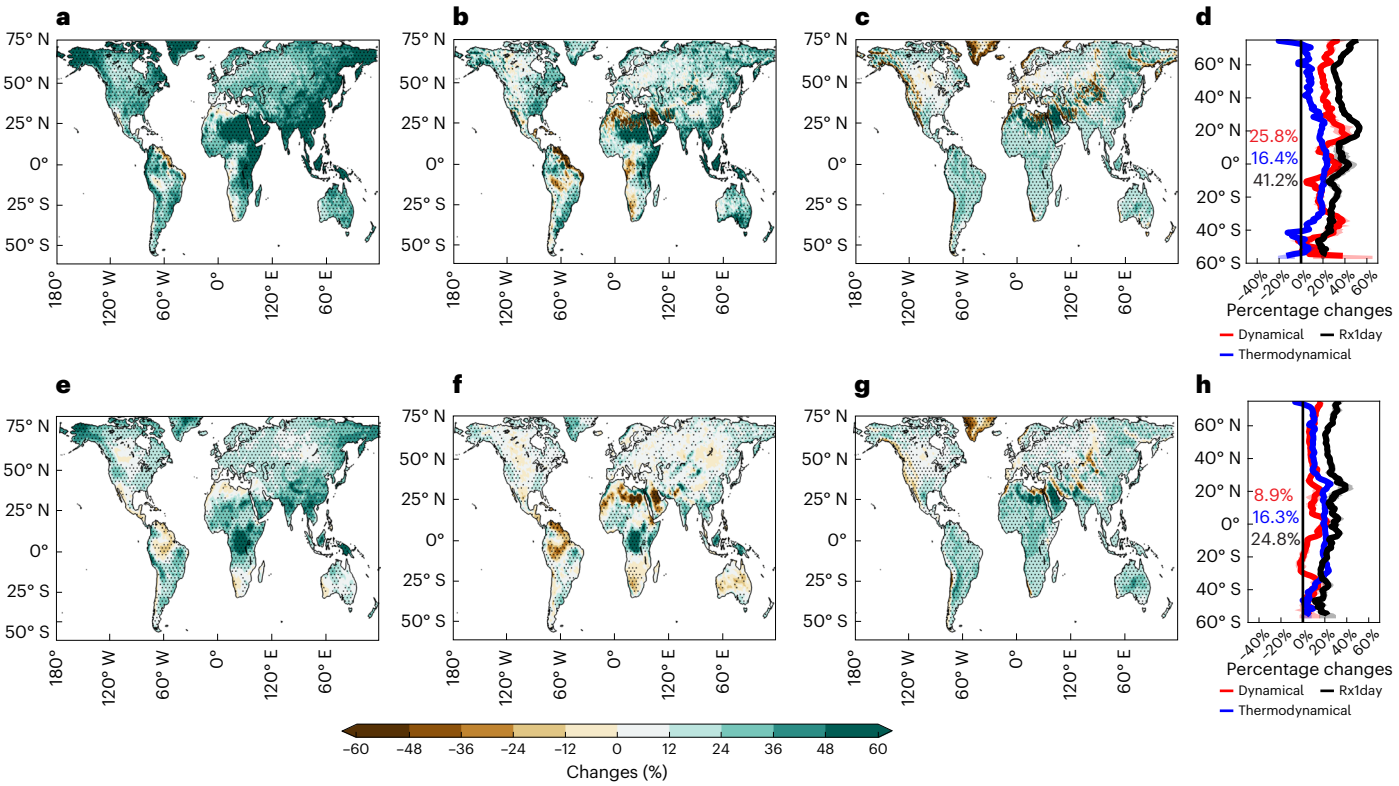


Fig. 5 | Projected fractional changes in Rx1day and their dynamical versus thermodynamical drivers. a–**c**, Percentage changes of (**a**) total Rx1day (mm/day), (**b**) changes driven by dynamical tendency and (**c**) driven by thermodynamical tendency over global land averaged over 2071–2100, relative to the 1981–2010 period in CESM-HR. Stippling indicates regions where three-quarters of the ensemble members agree on the sign of the ensemble mean. **d**, Zonally averaged Rx1day changes (41.2%) driven by dynamical (25.8%) and thermodynamical (16.4%) tendency across global land. Shading represents the standard deviation across 10-member ensembles. **e**–**g**, As in **a** (**e**), **b** (**f**) and **c** (**g**), but for CESM-LR. **h**, Zonally averaged Rx1day percentage changes (24.8%) driven by dynamical (8.9%) and thermodynamical (16.3%) tendency across global land for CESM-LR. The results indicate that the differences in projected future extreme precipitation between CESM-HR and CESM-LR are primarily driven by amplified atmospheric dynamical contributions, while thermodynamic contributions remain remarkably similar.

Tracking phenomena within the 24-h window of each annual Rx1day event during 2001–2020 shows that MCSs occur most often with fronts and ARs during Rx1day events over the SEUS, suggesting that multifeature interactions drive MCS formation in both observations and CESM-HR. However, CESM-LR largely misses these crucial interactions, leading to the near absence of MCSs, which contributes to the unrealistically weak Rx1day of CESM-LR. This finding that multiscale interactions generate MCSs and drive extreme precipitation holds in other regions of intense extreme precipitation, such as the classic Monsoon region (CMR; ref. 44) and the La Plata Basin (LPB).

Projected changes in extreme precipitation

The enhanced capabilities of CESM-HR substantially boost confidence in its projections of future extreme precipitation changes. To demonstrate the impact of model resolution, we compare Rx1day projections for the late twenty-first century (2071–2100) relative to recent decades (1981–2010) in CESM-HR and CESM-LR under RCP 8.5 (Fig. 4a,e). Both models project increased Rx1day across regions with intense extreme precipitation, including SEUS, CMR and LPB, but CESM-HR shows a larger increase, with a global-land-averaged Rx1day increase of 16.6 mm d^{-1} (from 40.3 mm d^{-1} to 56.9 mm d^{-1}), ~2.5 times greater than the 6.4 mm d^{-1} (from 25.8 mm d^{-1} to 32.2 mm d^{-1}) increase of CESM-LR. To further evaluate projected future changes in CESM-HR and CESM-LR, while accounting for differences in the historical climatologies of the model, we also calculated the fractional changes relative to the historical baseline (Fig. 5). Projected fractional changes are also higher almost everywhere in CESM-HR (Fig. 5a,e), with a global-land-averaged Rx1day increase of 41.2% in CESM-HR compared with 24.8% in CESM-LR by the end of the century (66.1% stronger). Regional differences are more striking: CESM-HR projects an increase exceeding 30 mm d^{-1} in parts of the CMR, five times the 6 mm d^{-1} increase projected by CESM-LR (corresponding to a 60% increase in CESM-HR versus 28% in CESM-LR). Over the contiguous United States (CONUS), CESM-LR projects a 4 mm d^{-1} increase for both the west coast and southeast, while CESM-HR projects a similar increase for the west coast but nearly triples it for the southeast, reaching over 12 mm d^{-1} (corresponding to a 30% increase in CESM-HR versus 12% in CESM-LR). The sharper Rx1day increase shown by CESM-HR from midlatitudes to subtropics and tropics aligns with its improved representation of MCSs, which drive extreme precipitation but are under-represented in CESM-LR.

The ensemble approach enables robust quantification of uncertainty arising from climate internal variability. Analysis of the ten-member CESM-HR ensemble reveals consistent projected changes across all members, with a global-land-averaged ensemble spread (standard deviation across ensemble members) of 6.1 mm d^{-1} and signal-to-noise ratio (SNR) of 2.7 (Supplementary Fig. 4). Notably, large projected increases in extreme precipitation exceeding the global mean of 16.6 mm d^{-1} over SEUS, CMR and LPB surpass the 95% SNR confidence level (Extended Data Fig. 4), highlighting the robustness of the projections for CESM-HR.

A moisture budget analysis highlights key differences in drivers of future extreme precipitation (Methods). In CESM-HR, the dynamical tendency dominates over the thermodynamical tendency across most regions of intense extreme precipitation (Fig. 4b–d and Fig. 5b–d),

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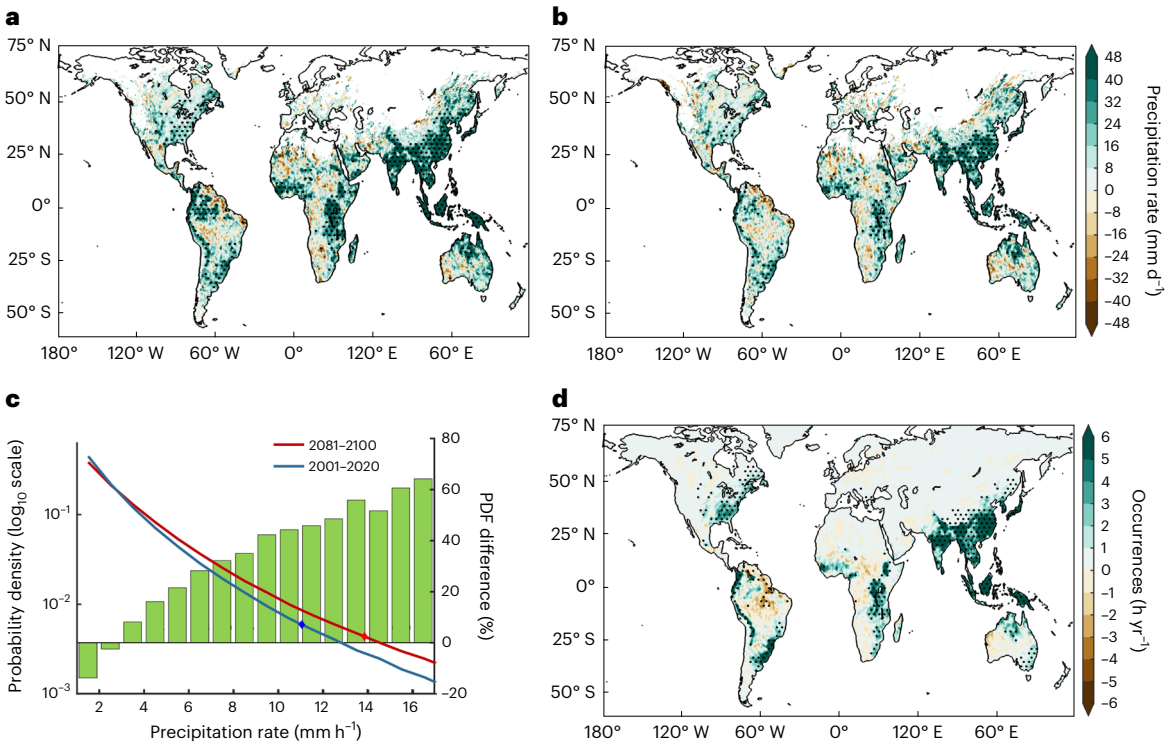


Fig. 6 | Projected changes in MCSs and associated precipitation. a, Projected future changes of Rx1day (mm d^{-1}) averaged over the future period (2081–2100) relative to the historical period (2001–2020) in CESM-HR ensemble member no. 10. The resulting pattern closely resembles the CESM-HR ensemble-mean projection of Rx1day changes shown in Fig. 4a. **b**, Similar to **a**, but only for MCSs-associated precipitation differences during Rx1day events. **c**, PDFs of CESM-HR simulated global-land-area MCSs-associated precipitation rates (mm h^{-1}) on a logarithmic scale (left vertical axis) for the historical (2001–2020) (blue) and future (2081–2100) (red) periods. The blue dot represents the historical period 99th percentile value (11.03 mm h^{-1}) and the red dot represents the future period 99th percentile value (13.82 mm h^{-1}). The green bars show fractional changes (right vertical axis) of the future period PDF relative to the historical period, with an increase of more than 60% for the most extreme MCSs in the future. **d**, Projected future changes in extreme MCS occurrences (h yr^{-1}) with precipitation rates exceeding the 99th percentile historical value. Stippling denotes regions where results exceed the 95% confidence level, as determined by two-sample *t*-test.

suggesting that future intensifications arise from both moisture and dynamics changes. In contrast, CESM-LR attributes Rx1day increases primarily to thermodynamical effects globally, except in the tropics where dynamical effects also contribute (Fig. 4f–h and Fig. 5f–h). The projected fractional changes attributed to thermodynamical contributions show similar spatial patterns in both CESM-HR and CESM-LR, with global-land-averaged values of 16.4% and 16.3%, respectively. However, the dynamical contribution to extreme precipitation fractional change is ~2.9 times stronger in CESM-HR (25.8%) than in CESM-LR (8.9%). These results indicate that the differences in future extreme precipitation projections between CESM-HR and CESM-LR are primarily driven by amplified atmospheric dynamical contributions, particularly enhanced mesoscale moisture convergence, while thermodynamical contributions from temperature-driven moisture increases remain remarkably similar. This finding is consistent in theoretical studies and regional cloud-resolving simulations[12,45], which also emphasize the key role of dynamical amplification in future extreme precipitation under global warming.

The stronger dominance of dynamical tendency in CESM-HR is closely tied to the increased occurrence of extreme precipitating MCSs under warmer conditions. Tracking MCSs and computing the concurrent extreme precipitation in present and future periods show that MCS-related extreme precipitation increases substantially contribute to the overall future increase (Fig. 6a,b), with MCS occurrences shifting towards more extremes in CESM-HR (Fig. 6c,d), an effect absent in CESM-LR. This future increase in extreme precipitating MCSs aligns with regional storm-resolving (~4 km) simulations over CONUS[12]. While not storm-resolving, our global simulations support regional model findings and extend projected MCS-driven extreme precipitation increases to a global scale.

High-resolution ensemble simulations at 10–25-km resolution offer promising improvements in simulating extreme precipitation during the historical period and enable more physically grounded future projections with better-constrained uncertainties. While unable to explicitly resolve clouds and deep convection, these simulations markedly outperform standard low-resolution CMIP simulations, more accurately replicate observed extreme precipitation statistics by improving representation of mesoscale atmospheric phenomena, such as MCSs and the multiscale interactions essential for capturing extreme events, underscoring the value of high-resolution models in enhancing the credibility of climate projections.

Building on these advancements, we advocate for the development of multimodel ensembles at 10–25-km resolution to advance global high-resolution climate modelling. Such efforts would enable robust intermodel comparisons, deepen understanding of extreme precipitation dynamics and produce more reliable projections critical for climate adaptation and mitigation. High-resolution datasets would also facilitate applications of AI and machine learning in climate research[46–48]. Because extreme precipitation events response to climate change is probably governed by non-stationary and complex processes, historical data alone are insufficient for training AI models intended for future applications. High-resolution simulations offer more realistic representations of extreme precipitation events, enabling AI tools to learn relationships relevant to a warming climate.

Despite their promise, high-resolution models still face challenges in simulating some observed extreme precipitation features.

For example, simulating subdaily extremes is generally more challenging than daily extremes[49] and CESM-HR struggles with warm-season extremes over CONUS, missing key contributions from MCSs in the central USA, which are vital for accurate extreme precipitation projections[50]. Addressing these gaps will require global storm-resolving models to explicitly resolve deep convection[20,21]. While high-resolution models represent a major step forward, further advancements, particularly through storm-resolving approaches, are essential to provide more precise representations of the atmospheric processes driving extreme events and to deliver more reliable projections of their future changes.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41561-025-01859-1.

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Methods

Observational precipitation data

In this study, we use gridded precipitation data from IMERG V07[41] and HadEX3[51] for verification. IMERG V07 provides level 3 'final run' precipitation analyses at a high spatial resolution of 0.1° longitude by 0.1° latitude, with daily data spanning from 1998 afterwards and near-global coverage. The IMERG algorithm integrates observations from several satellite sensors to deliver high-resolution precipitation data, covering both rainfall and snowfall. The latest version, v.7, introduces improvements to enhance the accuracy and reliability of precipitation estimates. We selected IMERG because of its high spatial and temporal resolution, its robust calibration against ground-based observations and other satellite data, and its 30-min temporal resolution, which is critical for identifying and tracking MCSs.

HadEX3, which provides globally gridded, station-based indices of temperature and precipitation extremes at a spatial resolution of 1.875° longitude by 1.25° latitude, covering the period from 1901 to 2018, is also used. It is derived from approximately 7,000 temperature and 17,000 precipitation stations worldwide and includes 29 indices, including the annual maximum daily precipitation. In HadEX3, extreme indices are first computed at individual stations and then interpolated onto the global grid using the angular distance weighting method. We selected this dataset for its long-term record and near-global coverage, which make it well-suited for assessing trends and variability in extreme precipitation at global and continental-scale, including Europe, Southeast Asia, North and South America, Africa and other regions.

Models: CESM simulations

We conducted ensembles of CESM-HR and CESM-LR historical-and-future transient climate simulations using CESM v.1.3 (CESM1.3) with many additional modifications and improvements. CESM1.3 consists of the community atmosphere model v.5 (CAM5) with the spectral element dynamical core for the atmospheric component, the parallel ocean program v.2 (POP2) for the ocean component, the community ice code v.4 (CICE4) for sea ice and the community land model v.4 (CLM4) for land.

CESM-HR has a nominal horizontal resolution of 0.25° for the atmosphere and land components and 0.1° for the ocean and sea-ice components (~10 km at the Equator down to ~4 km at high-latitudes), while CESM-LR has a nominal horizontal resolution of 1° for all components. Both CESM-HR and CESM-LR simulations include a 500-yr pre-industrial control climate simulation (PI-CNTL) and a historical-and-future transient climate simulation (HF-TNST) covering the period from 1850 to 2100 (ref. 29 gives details). PI-CNTL used perpetual 1850 climate forcing, while the HF-TNST branched from PI-CNTL at year 250, incorporating observed climate forcing through 2005 and following the 'high-emissions' RCP 8.5 and 'medium–high-emissions' RCP 6.0 scenarios from 2006 to 2100 in accordance with the protocol of CMIP5.

Additionally, nine HF-TNST ensemble members were generated by branching from the 1850–2100 HF-TNST simulation in 1920. These ensemble members were produced by applying round-off perturbations to the atmospheric potential temperature initial conditions. Thus, we have two ten-member HF-TNST ensembles, each under RCP 8.5 and RCP 6.0, for both CESM-HR and CESM-LR. All analyses in this study are based on the HF-TNST ensemble under RCP 8.5.

All CESM-HR simulation datasets are publicly available on the MESACLIP (https://project.cgd.ucar.edu/projects/MESACLIP/) and iPOGS (https://project.cgd.ucar.edu/projects/iPOGS/) project pages through the NSF NCAR Geoscience Data Exchange (GDEX) platform. Refer to section on 'Data Availability' for more details.

Climate simulations from CMIP6 and CMIP6-HighResMIP archives

We compared daily extreme precipitation results from our simulations with models participating in CMIP6[52] and HighResMIP[24]. Specifically, we computed extreme precipitation from 35 CMIP6 model historical simulations, spanning the period 1850–2014. These simulations have a nominal horizontal resolution of 1° or coarser (Supplementary Table 1). Additionally, we computed extreme precipitation from nine HighResMIP model historical simulations, covering the 1950–2014 period. These simulations have a nominal horizontal resolution ranging from 0.25° to 0.50° (Supplementary Table 2).

Although CESM-HR is also included in the HighResMIP project, the dedicated HighResMIP simulation design differs from that of the ten-member HF-TNST ensemble, covering only the 1950–2050 period and following a specialized protocol. To ensure consistency in comparisons between high- and low-resolution simulations, we analysed extreme precipitation over the overlapping period of 1950–2014 across all simulations.

Metric of extreme precipitation and spatial interpolation methods

We used two widely recognized indices to measure extreme precipitation: R99p and Rx1day. R99p represents the annual average of daily precipitation above the 99th percentile (top 1%) of recorded wet days ($\geq$0.1 mm d^{-1}). We used 0.1 mm d^{-1} to define wet days, following a recent study[53]. Rx1day denotes the annual maximum daily precipitation.

We acknowledge that the choice of spatial interpolation method (conservative versus bilinear) and the sequence of operations (that is, whether extremes are computed before or after interpolation) can introduce uncertainties in extreme precipitation estimates[33]. However, there is currently no consensus in the literature on the optimal approach for comparing model-simulated extremes with observations[54]. For instance, the frequent rainfall observations on grids (FROGS) intercomparison project[55] applied bilinear interpolation to reanalysis products (which were produced by Earth system models), while satellite observations were regridded to a common 1° × 1° grid using conservative interpolation. Similarly, CMIP6 comparison studies[56,57] have used bilinear interpolation for comparing simulated extremes with observations and for projecting future changes. At the same time, other studies adopt conservative remapping in the model–observation comparisons[33,58–60].

To assess sensitivity to interpolation method and processing sequence, we conducted parallel analyses using both approaches. For historical validation, we conservatively regridded IMERG V07, CESM-HR, CESM-LR, HighResMIP and CMIP6 simulations to a common 1° × 1° grid before computing R99p and land-region PDFs (Fig. 1). For comparison, we also computed R99p on native grids and then bilinearly interpolated the results (Supplementary Fig. 1). Consistent with previous findings[25,28], conservative remapping tends to dampen extremes in both observed and modelled data. Nevertheless, our main conclusion that CESM-HR produces more realistic extreme precipitation than CESM-LR remains robust across interpolation methods. A physical explanation for this consistency is discussed in ref. 25.

For Rx1day evaluation, we used HadEX3 data, which are derived from station observations, although coverage varies as a result of changes in station availability. We included only grid cells with data available for at least half the study period. Because HadEX3 uses angular distance weighting for gridding, which is similar to bilinear interpolation, we used bilinear interpolation for comparing model output to HadEX3 (Extended Data Fig. 1). We also followed a recent methodology[61] to assess simulated historical trends.

For future projections, CESM-HR and CESM-LR are compared using conservative interpolation. To ensure that differences in future projections are not merely artifacts of historical performance, we computed fractional changes in Rx1day (Fig. 5). We note that CESM-HR and CESM-LR exhibit very similar global warming rates[29].

To further investigate the physical sources of extreme precipitation during Rx1day events, we decomposed total precipitation into parameterized (from subgrid cumulus parameterization, labelled prc in CMIP6) and resolved (from large-scale processes) components

and quantified their contributions to each Rx1day event in CESM-HR and CESM-LR.

Moisture budget analysis

The moisture budget equation can be written as:

$$P-E=-\frac{1}{g}\nabla\cdot\int_0^{p_s}\mathbf{u}q\mathrm{d}p-\frac{1}{g}\frac{\partial}{\partial t}\left(\int_0^{p_s}q\mathrm{d}p\right),$$

where P and E represents precipitation and evaporation, respectively, $\mathbf{u}$ is the horizontal wind vector, q is the water vapour mixing ratio, p is air pressure, p_s is surface pressure and g is the gravitational acceleration. By discretizing the equation and defining the mass-weighted vertical integral of a variable X as $\{X\}=1/g\sum_k(X_k dp_k)$, where X_k is the value of X at the kth model level and $\mathrm{d}p_k$ is the pressure difference across this level, we can express the equation as:

$$P-E=-\nabla\cdot\{\mathbf{u}q\}-\frac{\partial}{\partial t}\{q\}=-\{q\nabla\cdot\mathbf{u}\}-\{\mathbf{u}\cdot\nabla q\}-\frac{\partial}{\partial t}\{q\}.$$

Thus, at a given grid point, $P-E$ is balanced by the column-integrated product of moisture q and dry flow convergence $\nabla\cdot\mathbf{u}$ (denoted M_c; $-\{q\nabla\cdot\mathbf{u}\}$), moisture advection (denoted M_a; $-\{\mathbf{u}\cdot\nabla q\}$) and local rate of change (tendency) of moisture (denoted M_t; $\frac{\partial}{\partial t}\{q\}$).

Six-hourly CESM-HR and CESM-LR outputs were used for this moisture budget analysis, consistent with a previous study[13]. Daily mean data tend to underestimate moisture flux convergence, especially because of the influence of midlatitude transient eddies[36], which are crucial in analysing precipitation extremes. Owing to the unavailability of 6-hourly outputs for one ensemble member, we conducted the analysis with a nine-member ensemble average.

To ensure a fair comparison between CESM-HR and CESM-LR, all moisture budget terms were computed on their native grids and then interpolated onto a 1° × 1° grid. The tendency term was calculated with a forward time difference:

$$\frac{\partial}{\partial t}\{q\}\approx\frac{\{q\}_{t+\Delta t}-\{q\}_t}{\Delta t},$$

where Δt = 6 h. During extreme events such as Rx1day, the complete moisture budget can be expressed as:

$$p^e=E^e-\{q^e\nabla\cdot\mathbf{u}^e\}-\{\mathbf{u}^e\cdot\nabla q^e\}-\frac{\partial}{\partial t}\{q^e\},$$

where the superscript 'e' represents extreme precipitation and associated processes. We note that Rx1day events occur on different dates across grid cells. To account for this spatial and temporal variability, we identify, for each grid cell, the specific date on which the local Rx1day event occurs, resulting in 270 unique events (30 years × 9 ensemble members). We then extract the corresponding moisture budget terms on those dates and compute their averages to characterize the event-specific composites of each moisture budget term for both models.

For extreme precipitation events in both CESM-HR and CESM-LR simulations, substantial cancellation occurs between moisture advection M_a and moisture tendency M_t, while evaporation E is negligible, consistent with previous studies[13]. Thus, extreme precipitation is approximately governed by:

$$P^e\approx-\{q^e\nabla\cdot\mathbf{u}^e\}.$$

Defining $C^e=-\nabla\cdot\mathbf{u}^e$ as dry flow convergence during an Rx1day event, we can express the difference in extreme precipitation between CESM-HR and CESM-LR, ΔP^e, as,

$$\Delta P^e=P^e_{HR}-P^e_{LR}\approx\{q^e_{HR}C^e_{HR}\}-\{q^e_{LR}C^e_{LR}\}.$$

Using a decomposition method similar to that introduced by a previous study[13] to separate contributions from thermodynamical and dynamical tendencies, we write:

$$\begin{aligned}\Delta P^e&\approx\{(q^e_{LR}+\Delta q^e)(C^e_{LR}+\Delta C^e)\}-\{q^e_{LR}C^e_{LR}\}\\&=\{\Delta q^eC^e_{LR}\}+\{q^e_{LR}\Delta C^e\}+\{\Delta q^e\Delta C^e\},\end{aligned}$$

where $\Delta q^e=q^e_{HR}-q^e_{LR}$ and $\Delta C^e=C^e_{HR}-C^e_{LR}$ represent the differences in specific humidity and dry flow convergence between CESM-HR and CESM-LR during their respective Rx1day events, respectively.

The nonlinear term, $\Delta q^e\Delta C^e$, is not negligible in the moisture budget governing Rx1day differences between CESM-HR and CESM-LR, indicating a nonlinear interaction between ΔC^e and Δq^e. Physically, this may suggest that an increase in local dry flow convergence in CESM-HR relative to CESM-LR could induce a specific humidity difference, which could further feedback on the flow convergence difference. For simplicity, we combine the last two terms on the right-hand side of the above equation, yielding:

$$\Delta P^e\approx\{\Delta q^eC^e_{LR}\}+\{q^e_{HR}\Delta C^e\}=M_{\Delta q}+M_{\Delta C},$$

where $M_{\Delta q}=\{\Delta q^eC^e_{LR}\}$ represents Rx1day differences between CESM-HR and CESM-LR attributable to moisture differences Δq^e (thermodynamical tendency) and $M_{\Delta C}=\{q^e_{HR}\Delta C^e\}$ reflects Rx1day differences resulting from dry flow convergence differences ΔC^e, including the nonlinear interaction (dynamical tendency). The results show that the dynamical tendency $M_{\Delta C}$ primarily drives Rx1day differences between CESM-HR and CESM-LR, while the thermodynamical tendency $M_{\Delta q}$ is secondary.

Similar decompositions can be applied to evaluate the contributions from thermodynamical and dynamical tendencies to projected future changes in Rx1day (ref. 13). For a future climate simulation, we define Δq^e and ΔC^e as differences in specific humidity and dry flow convergence between future (2071–2100) and historical (1981–2010) periods:

$$q^e_{fut}=q^e_{hist}+\Delta q^e\text{and}C^e_{fut}=C^e_{hist}+\Delta C^e$$

The future change in extreme precipitation relative to the historical period is governed by:

$$\Delta P^e=P^e_{fut}-P^e_{hist}\approx\{q^e_{fut}C^e_{fut}\}-\{q^e_{hist}C^e_{hist}\}=\{\Delta q^eC_{hist}{}^e\}-\{q^e_{fut}\Delta C^e\}.$$

The ensemble-mean future changes in Rx1day (ΔP^e) projected by CESM-HR and CESM-LR are compared in Fig. 4, along with the contributions from the dynamical ($\{q^e_{fut}\Delta C^e\}$) and thermodynamical ($\{\Delta q^eC_{hist}{}^e\}$) tendencies. Figure 5 presents similar results for future projections but shown as percentage changes relative to the historical mean. We note that the dynamical tendency uses the future specific humidity, and thus includes the nonlinear effect that is strong in CESM-HR. Consistent with previous findings[13], CESM-LR shows more substantial contributions from the thermodynamical tendency in driving future Rx1day increase across global land areas. In contrast, CESM-HR reveals a much stronger dynamical tendency contribution, indicating that regional circulation changes play a substantial role in driving extreme precipitation changes under future warming. This finding is consistent with the theoretical study by ref. 45 using the column quasi-geostrophic framework, which also emphasizes the role of dynamical amplification in future extreme precipitation under a warming climate.

MOAAP algorithm and multiscale interactions

The MOAAP algorithm was developed to detect and track atmospheric phenomena, analysing their occurrence frequencies and interactions during extreme precipitation events[17]. This Python-based package

requires only a minimal set of input variables: 1-hourly integrated water vapour transport, surface pressure, geopotential height, temperature, zonal and meridional winds at selected pressure levels, precipitation and brightness temperature. Using MOAAP, we identified ten primary extreme precipitation-producing atmospheric phenomena—mid-level anticyclones, surface anticyclones, ARs, COLs, mid-level cyclones, surface cyclones, fronts, jets, MCSs and TCs—along with five types of equatorial waves—eastward-propagating inertia-gravity waves, Rossby waves, inertia-gravity waves, Kelvin waves and mixed Rossby-gravity waves. To ensure consistency, we applied the same thresholds and criteria across observations, CESM-HR and CESM-LR simulations. Detailed descriptions of the MOAAP algorithm and threshold settings can be found in section 2 of ref. 17.

The observed analysis in ref. 17 was based on hourly data from ERA5 reanalysis[40], IMERG V07 and GPM-MERGIR[42]. The results shown in Fig. 3 update those from ref. 17 using the latest IMERG V07 data. To align GPM with the ERA5 resolution, IMERG V07 and GPM-MERGIR were interpolated onto a 0.25° × 0.25° grid before applying the MOAAP algorithm to global observations from 2001 to 2020. Notably, MCSs were identified using only IMERG V07 and GPM-MERGIR precipitation and brightness temperature data, unaffected by potential ERA5 biases. MCS ice clouds were tracked by masking all grid cells with brightness temperatures ≤241 K and excluding features with cloud shields smaller than 40,000 km^2 lasting less than 4 h. Additionally, MCSs were required to exhibit overshooting tops (brightness temperatures below 215 K), peak precipitation rates over 15 mm h^{-1} and a minimum precipitation area of 5,000 km^2 at least once during their lifetime. These thresholds and methods are widely used in MCS studies[62,63].

The CESM-HR and CESM-LR ensemble datasets provide atmospheric output at a 6-hourly frequency, which is insufficient for tracking MCSs. To address this, we reran CESM-HR and CESM-LR simulations for the historical (2001–2020) and the future period (2081–2100) with hourly outputs, limited to one ensemble member per model because of data volume and computational constraints. Given the large number of tracked features within a single ensemble member, increasing the ensemble size is unlikely to notably impact the results. It is also worth noting that the observations represent a single realization of the climate system, similar in nature to a single ensemble member, which supports the validity of our approach. MOAAP was applied to the native CESM-HR and CESM-LR grids and tracked features were subjected to further statistical analyses.

Using MOAAP-identified atmospheric phenomena, we constructed scale diagrams to analyse the spatiotemporal characteristics of these features (Supplementary Table 3). Spatial scales were estimated by calculating the area of each phenomenon (assuming circular shapes), while temporal scales were determined on the basis of tracking duration. For example, both observed and CESM-HR simulated MCSs show an average lifespan of under one day and a median spatial extent of ~300 km, which increases to ~800 km in CESM-LR, indicating the inability of CESM-LR to realistically represent MCSs.

Occurrence frequency for each tracked phenomenon was calculated as the accumulated days per year that a tracked phenomenon was identified at a grid point.

To link the tracked atmospheric phenomena to extreme precipitation, we identified the most frequent phenomenon for each Rx1day event at each grid point by selecting the one with the highest average occurrence frequency across all events at that location.

We further analysed potential multiscale interactions among atmospheric phenomena during Rx1day events using the SEUS as a case study. The occurrence of each phenomenon was normalized as a percentage of the total occurrences for all tracked phenomena. For each Rx1day event at a grid point in the SEUS, we calculated the total occurrences of all phenomena within the 24-h window of the event, ranked them from highest to lowest and normalized each by the total. Area-averaged normalized frequencies across all SEUS grid points for each year (2001–2020) reveal the phenomena most frequently co-occurring with MCSs during annual Rx1day events. Averaging these frequencies over the 2001–2020 period highlights the phenomena that most commonly co-occur with MCSs in the SEUS. While this analysis does not require phenomena to occur simultaneously with MCSs, we consider those with the highest normalized frequencies alongside MCSs within the 24-h window of a Rx1day event as concurrent, recognizing that MCS formation is often tied to the environmental conditions shaped by these phenomena rather than their exact timing. All data supporting the findings of this study are publicly available in the repositories listed in the section on 'Data availability'.

Data availability

All data supporting the findings of this study are available from public repositories. Observed precipitation data from HadEX3 and IMERG V07 were downloaded from the Met Office HadEX3 portal (https://www.metoffice.gov.uk/hadobs/hadex3/) and GPM-IMERG dataset page of NASA (https://disc.gsfc.nasa.gov/datasets/GPM-3IMERGDF_07/summary), respectively. HighResMIP and CMIP6 model outputs are available via the ESGF node (https://esgf-node.llnl.gov/search/cmip6/). ECMWF ERA5 reanalysis data can be accessed at https://www.ecmwf.int/en/forecasts/dataset/ecmwf-reanalysis-v5. As part of this study, we are releasing the ten-member CESM-HR ensembles of historical-and-future transient climate simulations (1920–2100), which are available through the NSF NCAR Geoscience Data Exchange (GDEX) platform at https://gdex.ucar.edu/datasets/d651007/ (historical ensemble)[64], https://gdex.ucar.edu/datasets/d651008/ (RCP 6.0 future ensemble)[65] and https://gdex.ucar.edu/datasets/d651009/ (RCP 8.5 future ensemble)[66].

Code availability

The MOAAP code are available via Zenodo at https://doi.org/10.5281/zenodo.7561616 (ref. 67). The CESM codes are available via GitHub at https://github.com/ihesp/cesm/tree/ihesp-hires-master.

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Author contributions

P.C. and G.D. led the high-resolution modelling projects. P.C. and D.F. conceptualized this study and wrote the original paper. D.F. designed the research framework and led the scientific analysis. X.L. and X.W. assisted with the analysis. F.S.C., Q.Z., T.K. and N.R. performed the ensemble simulations. A.F.P. developed the MOAAP package and conducted the initial MOAAP analysis, while D.F. and F.S.C. performed further MOAAP analysis. G.D, J.B., F.S.C. and S.C.B. helped interpret the results. All authors contributed to improving the paper.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41561-025-01859-1.

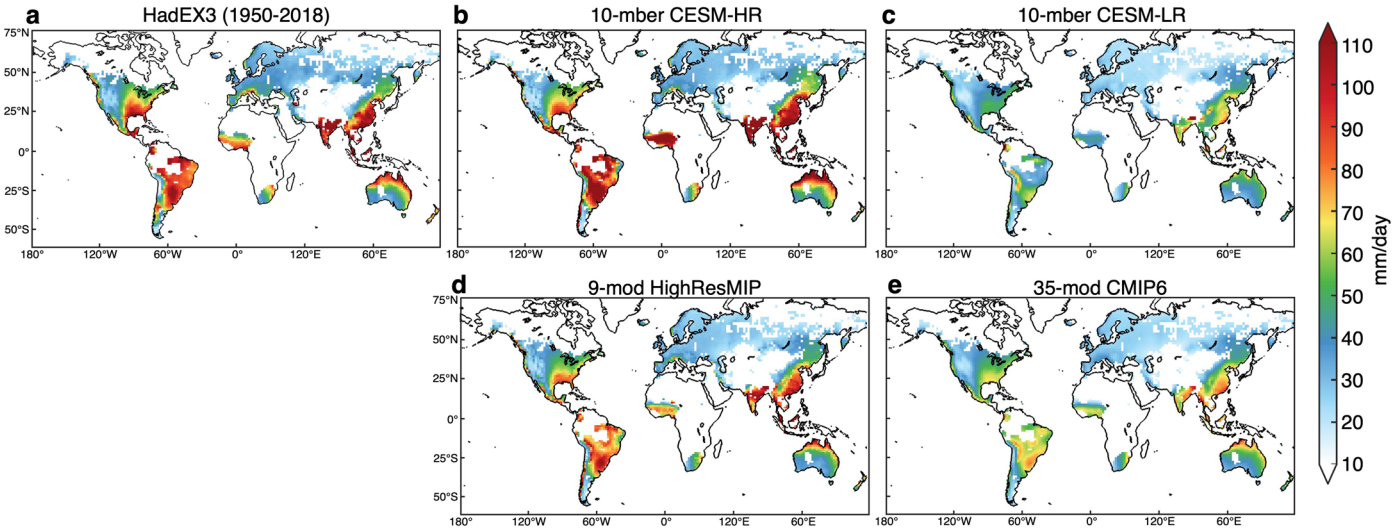
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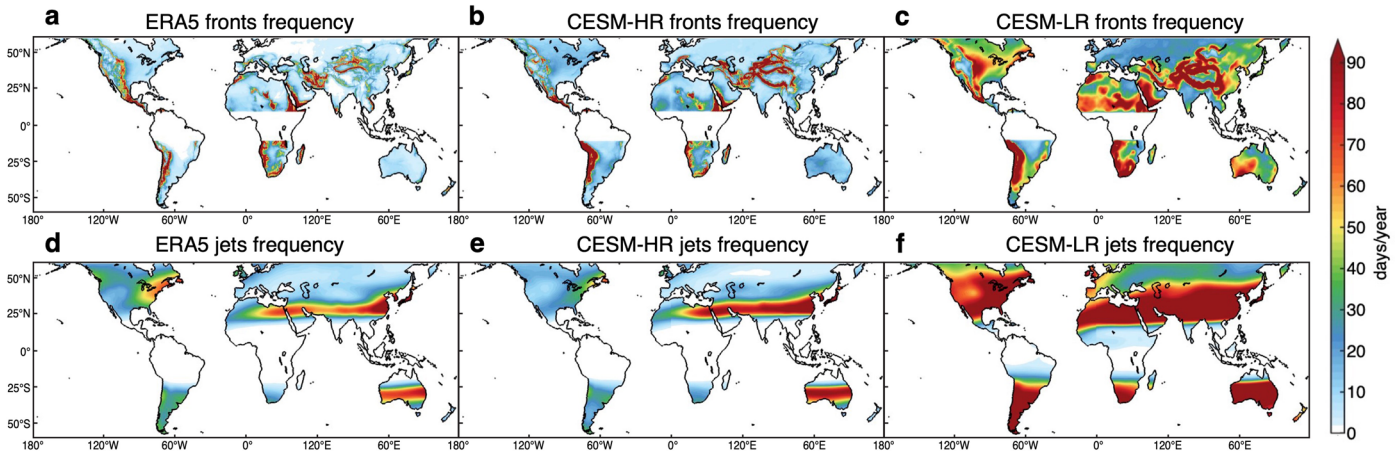
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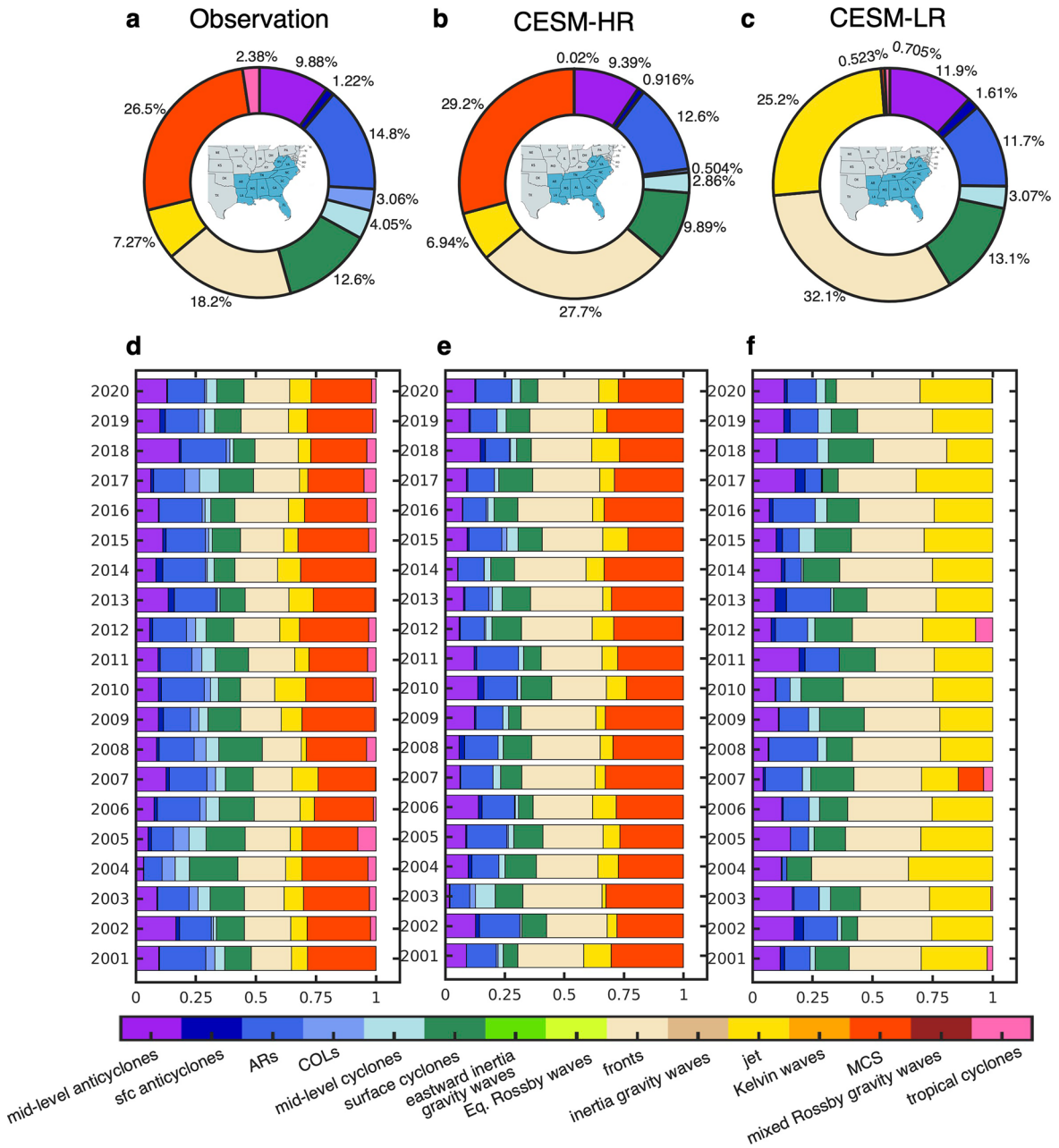
Extended Data Fig. 1 | Comparison of observed and simulated annual maximum daily precipitation. (**a**) Observed Rx1day (mm/day) averaged over the 1950–2018 period from the Hadley Centre Global Climate Extremes Index version 3 (HadEX3). Areas without shading indicate missing data where records cover less than half of the 1950-2018 period. (**b**), (**c**) As in (**a**), but for the 10-member ensemble mean of CESM-HR and CESM-LR, respectively. (**d**), (**e**) As in (**a**), but for the multi-model ensemble mean of 9 HighResMIP models and 35 CMIP6 models for the period of 1950-2014, respectively. To ensure consistency with the HadEX3, Rx1day are first computed on the model native grid and then regridded onto the HadEX3 grid using bilinear interpolation method. The global mean Rx1day is 61.3 mm/day, 66.8 mm/day, 38.8 mm/day, 53.0 mm/day, and 45.3 mm/day, for HadEX3, CESM-HR, CESM-LR, HighResMIP, and CMIP6, respectively.

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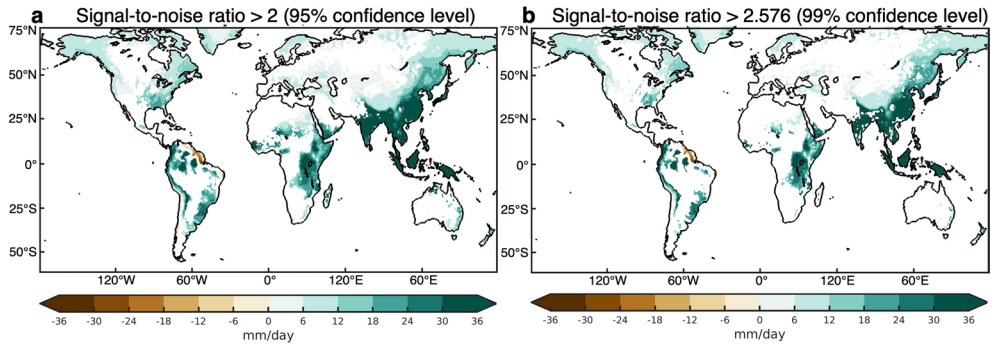
Extended Data Fig. 2 | Multi-Object Analysis of Atmospheric Phenomena (MOAAP) program tracked fronts and jets occurrence frequency. (**a**) Occurrence frequency (days/year) for MOAAP-tracked fronts from ERA5. (**b**), As in (**a**), but for tracked fronts from CESM-HR. (**c**) As in (**a**), but for tracked fronts from CESM-LR. (**d**), Occurrence frequency (days/year) for MOAAP-tracked jets from ERA5. (**e**) As in (**f**), but for tracked jets from CESM-HR. (**f**) As in (**d**), but for tracked jets from CESM-LR. To be consistent with tracked mesoscale convective systems shown in Fig. 3, the spatial maps are shown between 60°S and 60°N.

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Extended Data Fig. 3 | Normalized phenomenon frequency over Southeastern United States (SEUS). (**a**) Normalized occurrence frequency (%) of tracked atmospheric phenomena averaged over 2001-2020 for SEUS in observations, (**b**) As in (**a**), but for CESM-HR. (**c**) As in (**a**), but for CESM-LR. (**d**) Stacked bar plot of normalized occurrence frequency (%) of tracked phenomena over SEUS within the 24-hour window for each annual Rx1day event from 2001 to 2020 in observations. (**e**), As in (**d**), but for CESM-HR. (**f**) As in (**d**), but for CESM-LR. SEUS is shaded in blue on the map in (**a**-**c**).

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Extended Data Fig. 4 | Projected Rx1day future changes exceeding signal-to-noise thresholds. (**a**) Projected Rx1day changes (mm/day) for the future period (2071-2100) relative to the present period (1981-2010) over regions where the signal-to-noise ratio exceeds the 95% confidence level (SNR > 2), (**b**) As in a, but for regions where the signal-to-noise ratio exceeds the 99% confidence level (SNR > 2.576).